Chapter 4
Local Area network

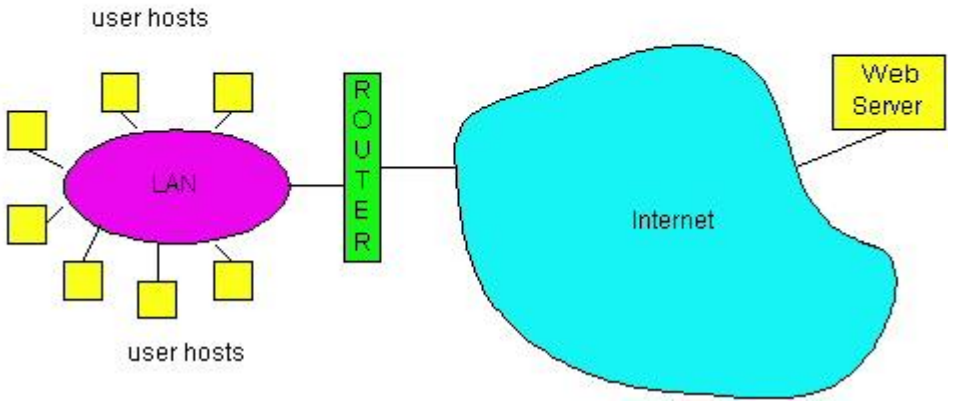
LAN technologies

Data link layer so far:

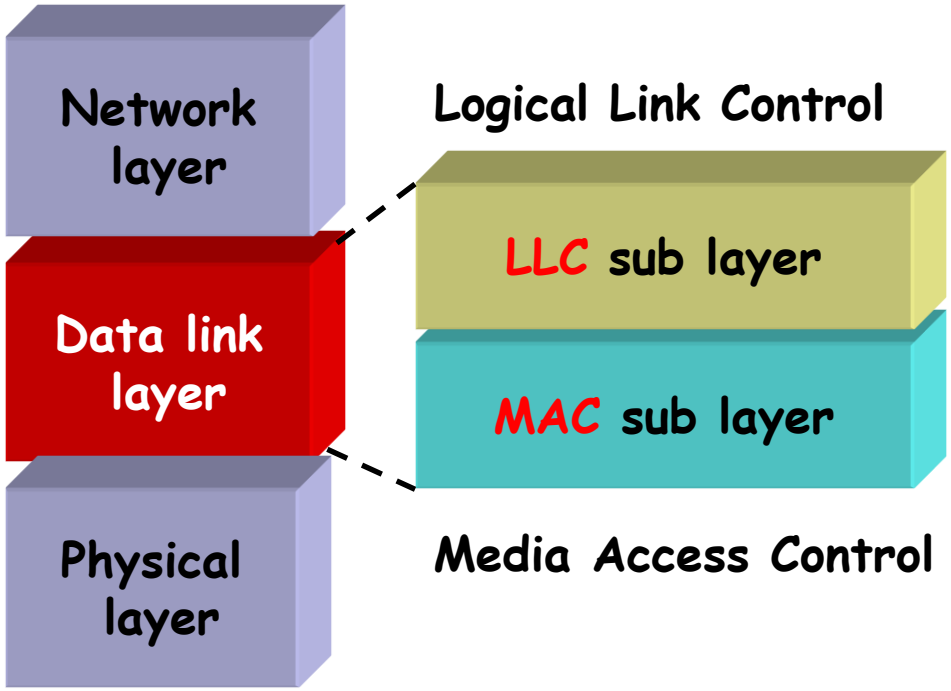
- services, error detection/correction, multiple access

Next: LAN technologies

- LAN model
- addressing
- Ethernet
- hubs, switches



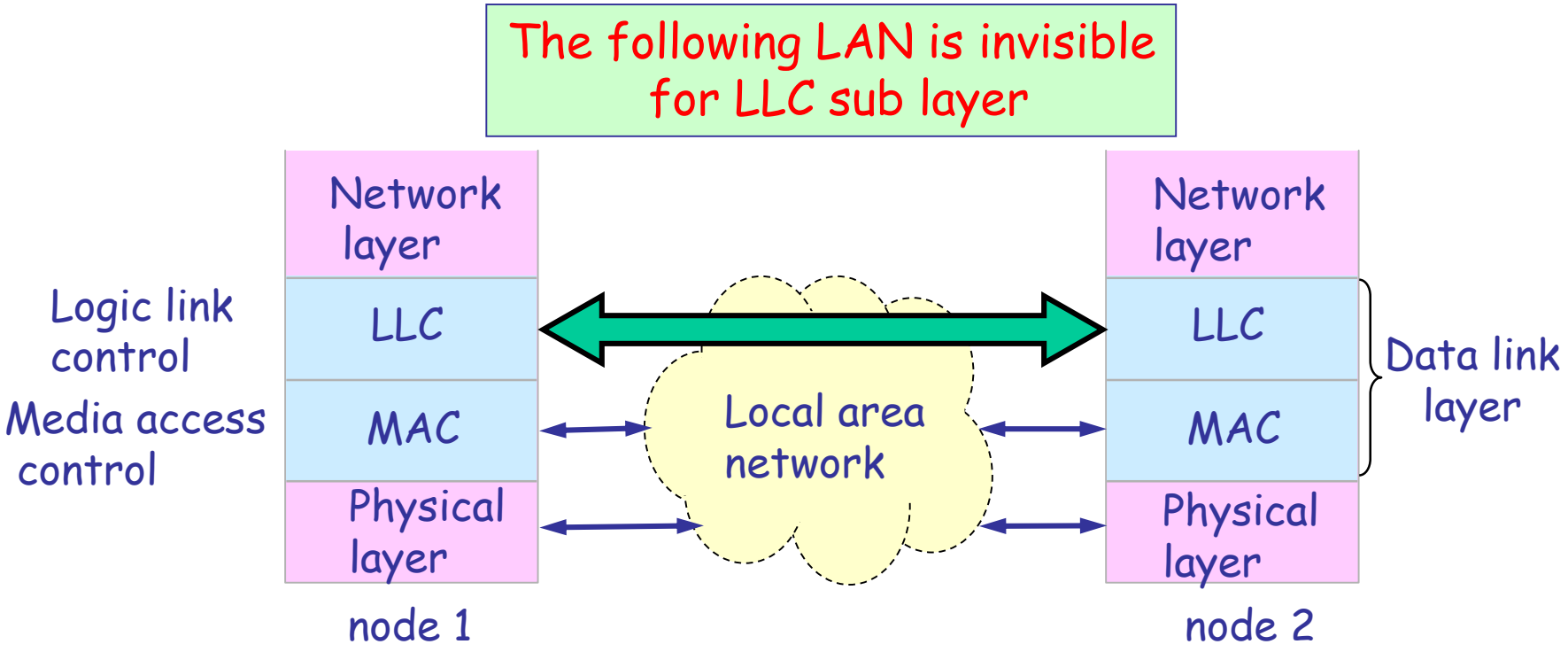
LAN model



Provide the connection environment to the upper level

Providing media access to the lower layer

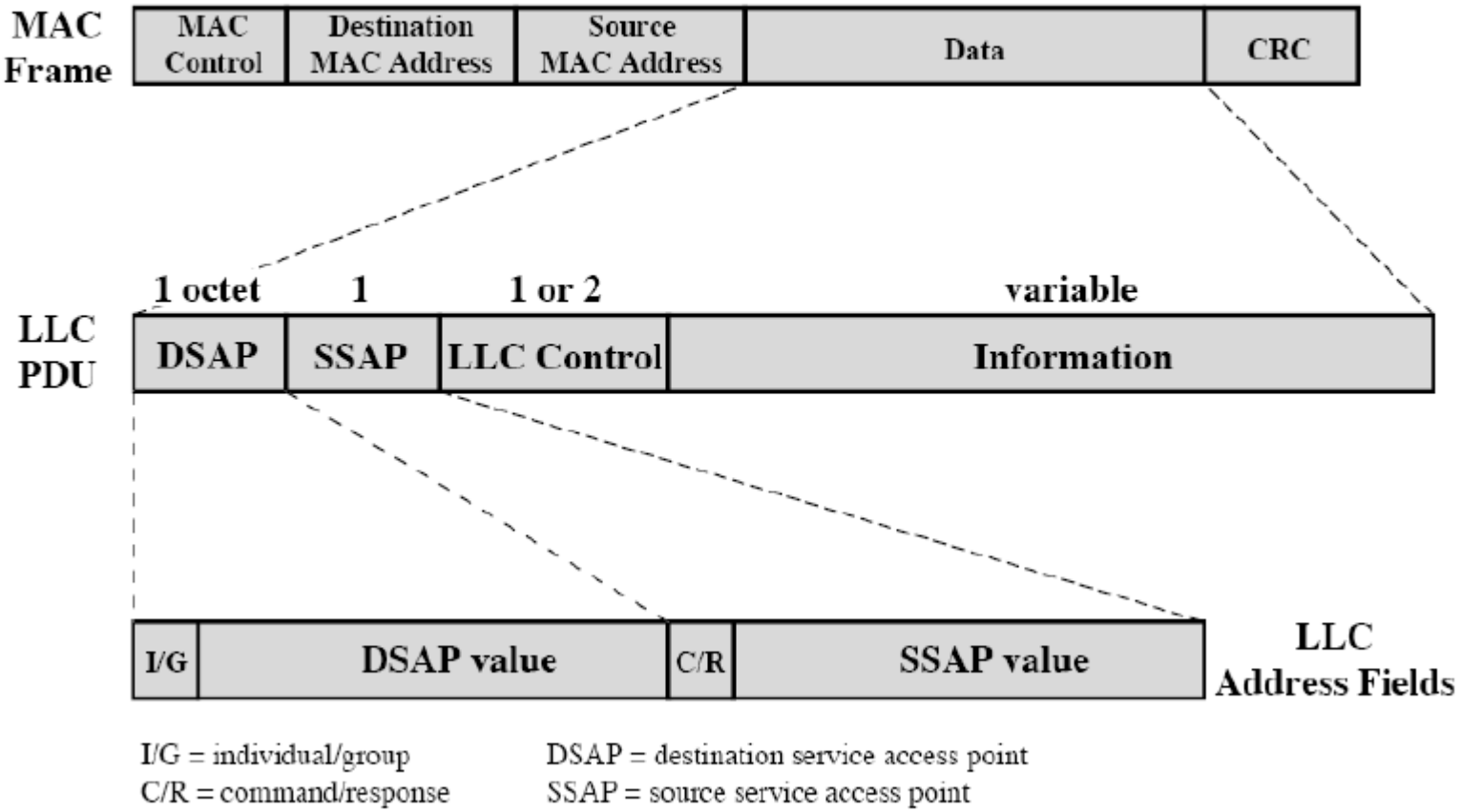
LAN model



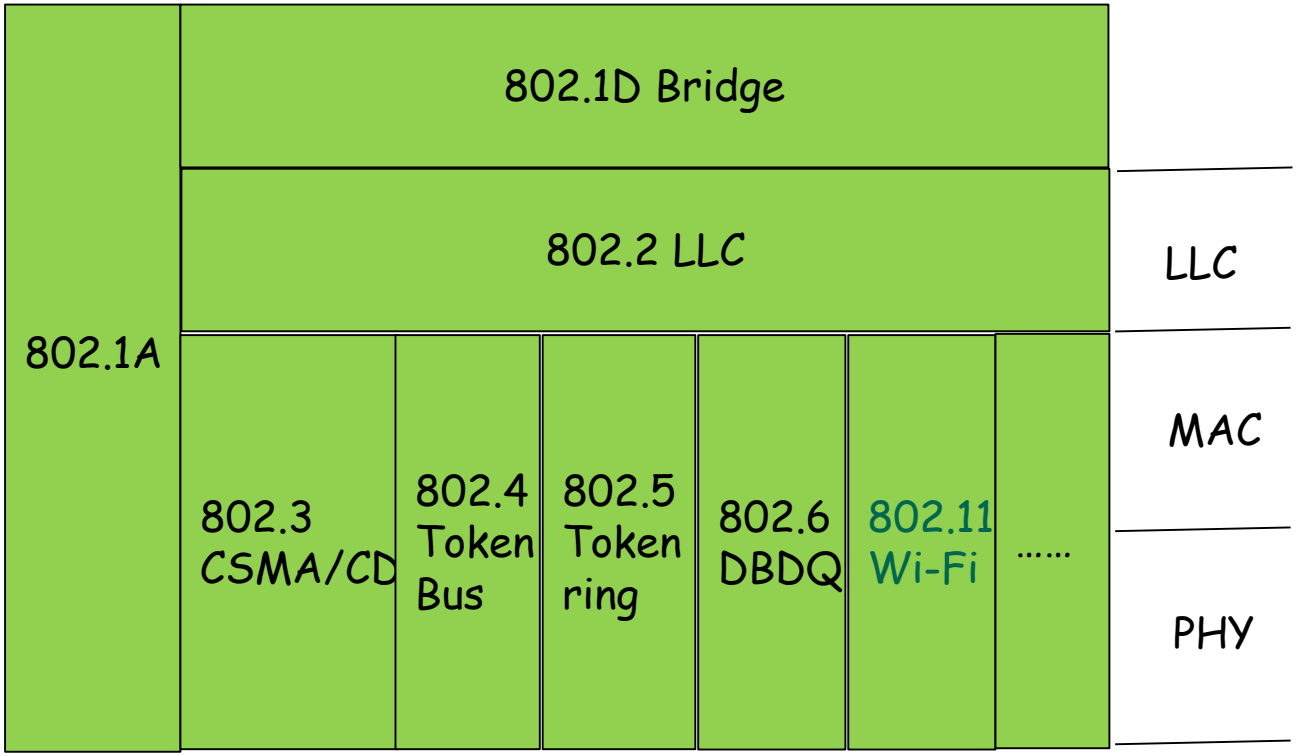
For the same LLC, several MAC options may be provided.

LLC and MAC

MAC Frame Format



IEEE 802 working group



Devices on a LAN are often interconnected using redundant links to improve resilience. This, however, may cause loops, which can consume network resources and break down the network.

802.1D implements the Spanning Tree Protocol, which builds a loop-free logical topology for LANs, thus blocking redundant links between any two network devices.

LAN Addresses

32-bit IP address:

- *network-layer* address
- used to get datagram to destination network

LAN (also known as MAC or physical) address:

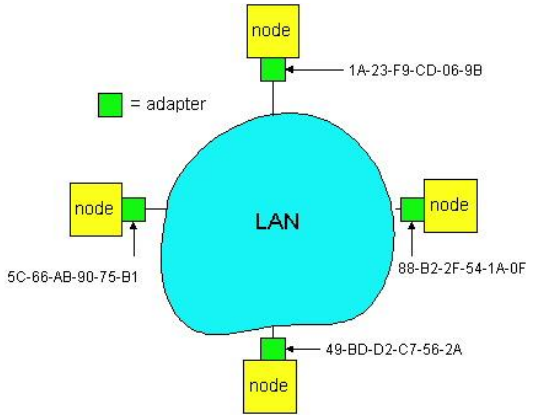
- unique identifier of a network adapter
- used to get datagram from one interface to another physically-connected interface (same network)
- 48 bit MAC address (for most LANs) burned in the adapter Read-Only Memory (ROM)

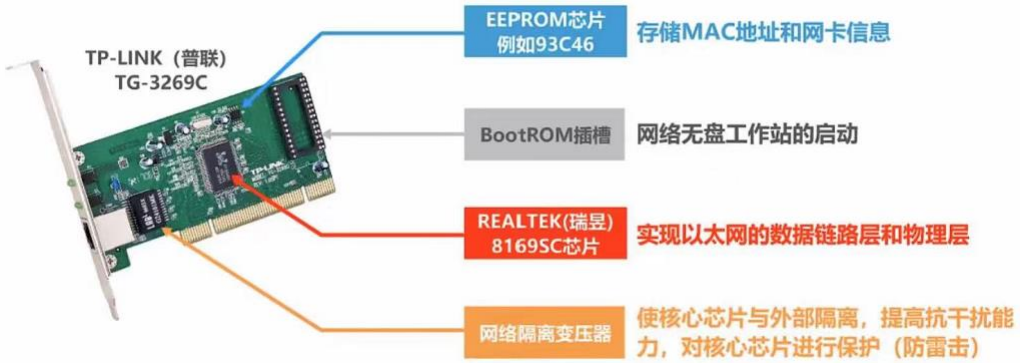
LAN Address (more)

- MAC address allocation administered by IEEE
- manufacturer buys portion of MAC address space (to assure uniqueness)
- Analogy:
  - (a) MAC address: like Social Security Number
  - (b) IP address: like postal address
- MAC flat address => portability
  - can move LAN card from one LAN to another
- IP hierarchical address NOT portable
  - depends on network to which one attaches

LAN Address

Each adapter on LAN has unique LAN address





IEEE 802局域网的MAC地址格式

扩展的唯一标识符EUI
EUI-48

组织唯一标识符OUI（由IEEE的注册管理机构分配）								网络接口标识符（由获得OUI的厂商自行随意分配）							
第一字节								第二字节							
b7	b6	b5	b4	b3	b2	b1	b0	b7	b6	b5	b4	b3	b2	b1	b0
十六进制	X		X		X		X		X		X		X		X	

标准表示法：	XX-XX-XX-XX-XX-XX	Windows	例如：00-0C-CF-93-8C-92
其他表示法：	XX:XX:XX:XX:XX:XX		例如：00:0C:CF:93:8C:92
	XXXX.XXXX.XXXX	Cisco Packet Tracer	例如：000C.CF93.8C92

MAC Addresses



第一字节的b1位	第一字节的b0位	MAC地址类型	地址数量占比	总地址数量
0	0	全球管理 单播地址 厂商生产网络设备（网卡，交换机，路由器）时固化	1/4	2^{48}=281,474,976,710,656 （二百八十多万亿）
	1	全球管理 多播地址 标准网络设备所支持的多播地址，用于特定功能	1/4	
1	0	本地管理 单播地址 由网络管理员分配，覆盖网络接口的全球管理单播地址	1/4	
	1	本地管理 多播地址 用户对主机进行软件配置，以表明其属于哪些多播组 注意：剩余46位全为1时，就是广播地址FF-FF-FF-FF-FF-FF	1/4	

https://standards-oui.ieee.org/oui/oui.txt
MAC地址查询 - https://mac.bmcx.com/

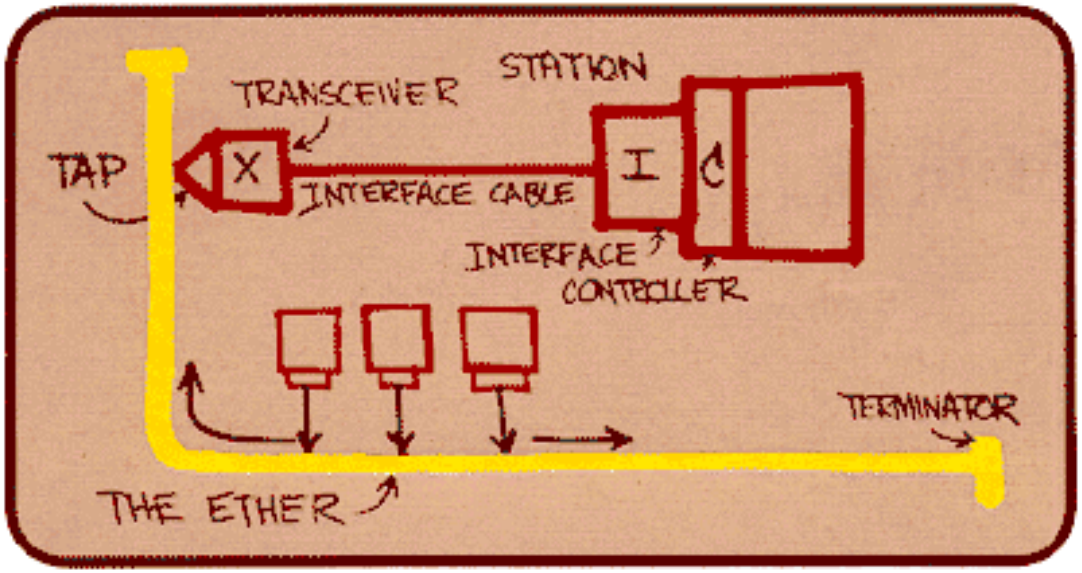
What is the random MAC address technology?

Privacy feature that uses a different, temporary MAC address for each LAN to prevent devices from being tracked across different networks. By replacing a device's unique hardware address with a random one, it makes it difficult for observers to monitor user movements & behaviors, especially across different public Wi-Fi hotspots. It is a default feature in many modern operating systems like iOS, Android, and macOS

Ethernet

"dominant" LAN technology:

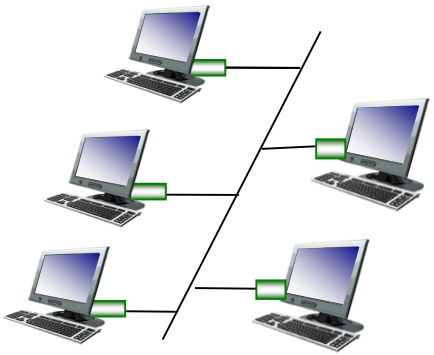
- (early version of) Fast Ethernet 100Mbs
- Modern standards: Gigabit Ethernet (1 Gbps), 10 Gigabit Ethernet (10 Gbps), and even faster speeds (25 Gbps, 40 Gbps, 50 Gbps, and 100 Gbps)
- first widely used LAN technology



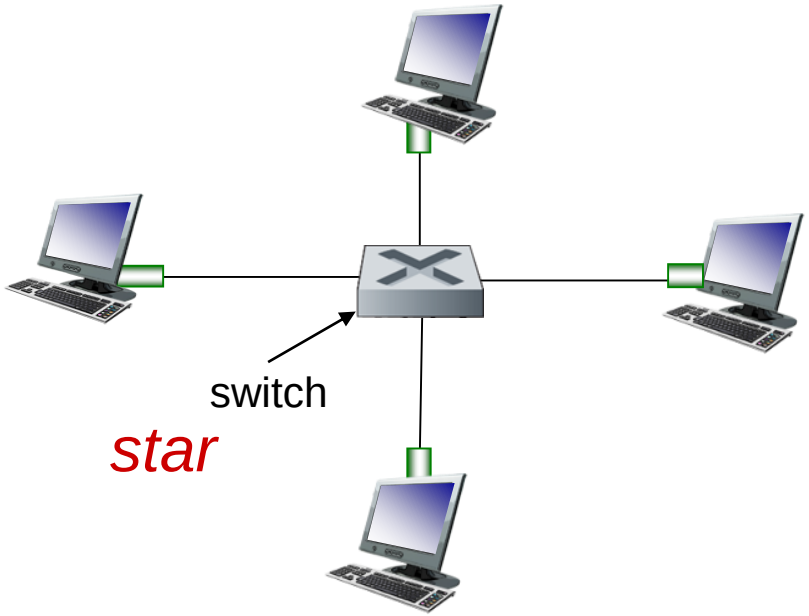
Metcalfe's Etheret sketch

Ethernet: physical topology

- *bus:* popular through mid 90s
  - all nodes in same collision domain (can collide with each other)
- *star:* prevails today (modern Ethernet)
  - active *switch* in center
  - each "spoke" runs a (separate) Ethernet protocol (nodes do not collide with each other)



bus: coaxial cable



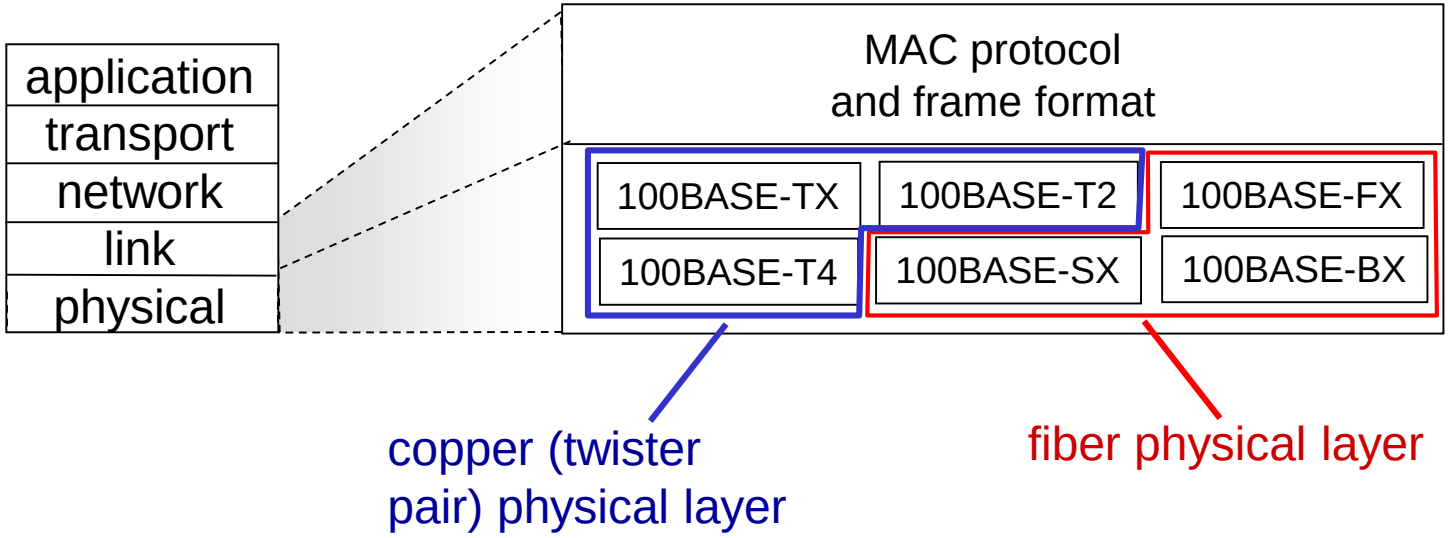
star

Ethernet: unreliable, connectionless

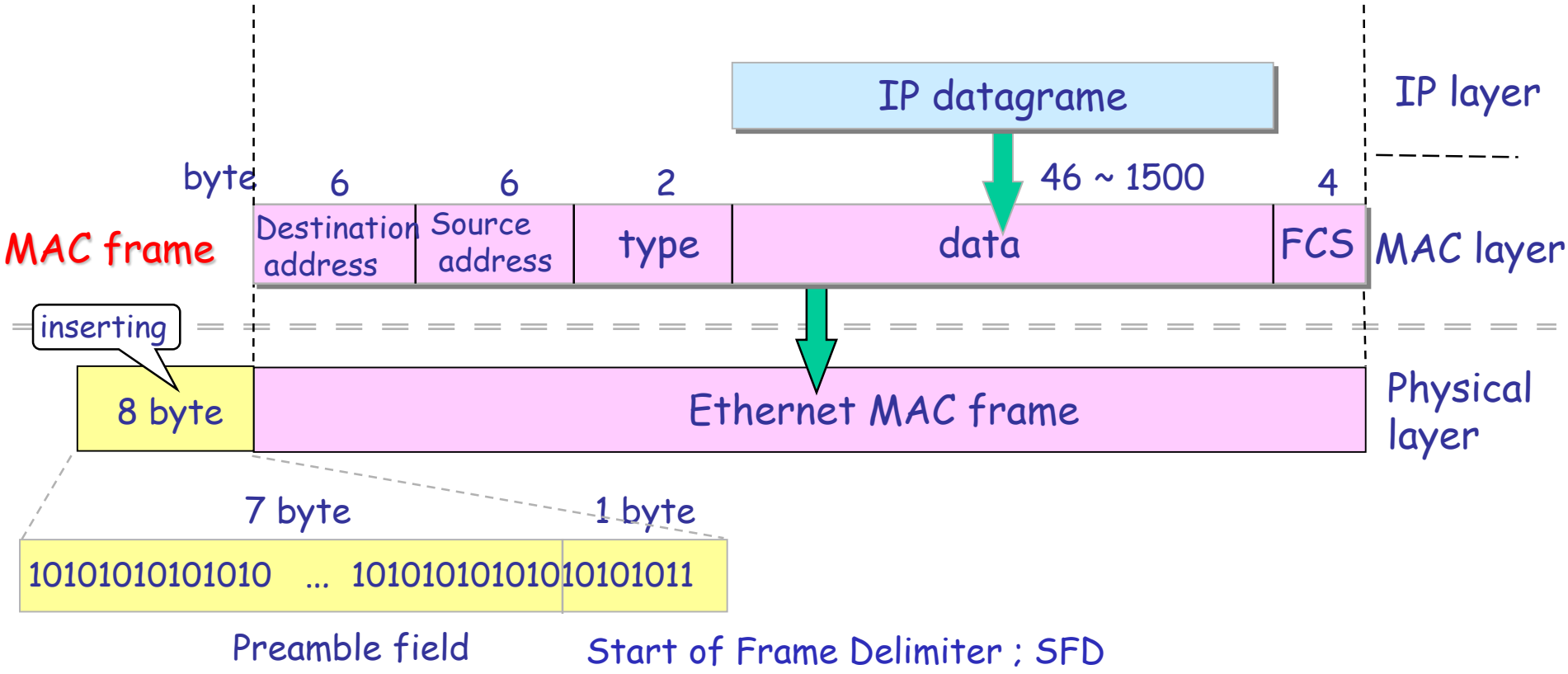
- *connectionless:* no handshaking between sending and receiving network interface cards (NICs)
- *unreliable:* receiving NIC doesn't send acks or negative acks (nacks) to sending NIC
  - data in dropped frames recovered only if initial sender uses higher layer recovery data techniques (e.g., TCP), otherwise dropped data lost
- Ethernet's MAC protocol: unslotted *CSMA/CD with binary backoff*

802.3 Ethernet standards: link & physical layers

- *many* different Ethernet standards
  - common MAC protocol and frame format
  - different speeds: 10 Mbps, 100 Mbps, 1Gbps, 10 Gbps, 25 Gbps, 40 Gbps, 50 Gbps, 100 Gbps
  - different physical layer media: fiber, cable



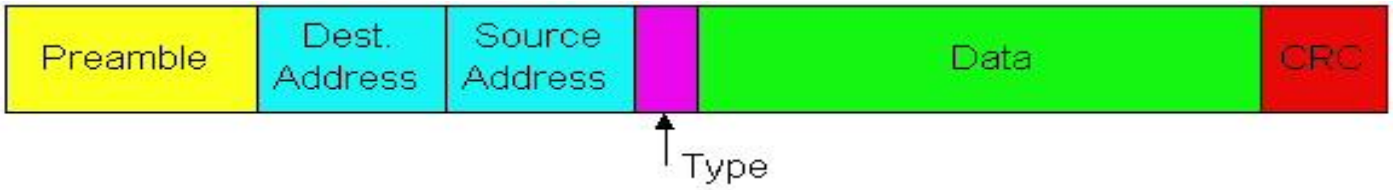
Ethernet Frame Structure



FCS (Frame Check Sequence): a 32-bit error-detecting code, calculated using a Cyclic Redundancy Check (CRC) algorithm

Ethernet Frame Structure

Sending adapter encapsulates IP datagram (or other network layer protocol packet) in Ethernet frame

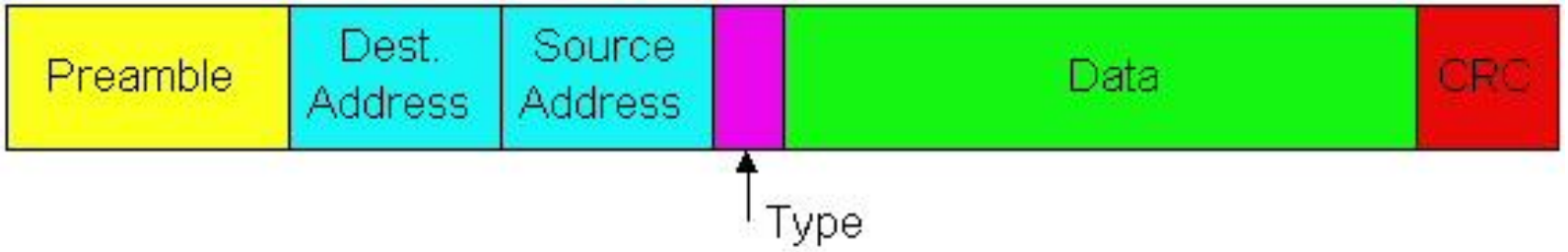


Preamble:

- 7 bytes with pattern 10101010 followed by one byte with pattern 10101011
- used to synchronize receiver, sender clock rates

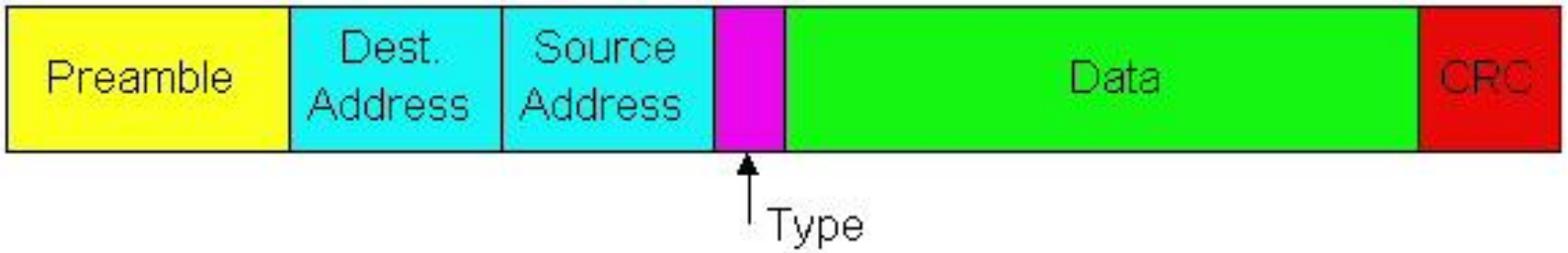
Ethernet Frame Structure (more)

- Addresses: 6 bytes, frame is received by all adapters on a LAN and dropped if address does not match
- Type: 2 bytes, indicates the higher layer protocol, mostly IP but others may be supported such as Novell IPX and AppleTalk)
- CRC: 4 bytes, checked at receiver, if error is detected, the frame is simply dropped



Ethernet Frame Structure (more)

- Data: 46~1500 bytes
- Minimum frame length: 64 bytes, why? (contention period 2τ is 51.2 μs for IEEE 802.3, R=10Mbps)
- Maximum frame length: 1518 bytes, why?
  It ensured that one station didn't block others on the shared medium for too long and also allowed NICs to maintain a small amount of then-expensive buffer memory



Exercise-1

- In a LAN using CSMA/CD protocol, the transmission medium is a complete cable, the transmission rate is 1 Gbps, and the signal propagation rate in the cable is 200,000 km/s. if the minimum data frame length is reduced by 800 bits, the distance between the farthest two stations needs to be at least

(1) increased by 160m (2) increased by 80m
(3) reduced by 160m (4) reduced by 80m

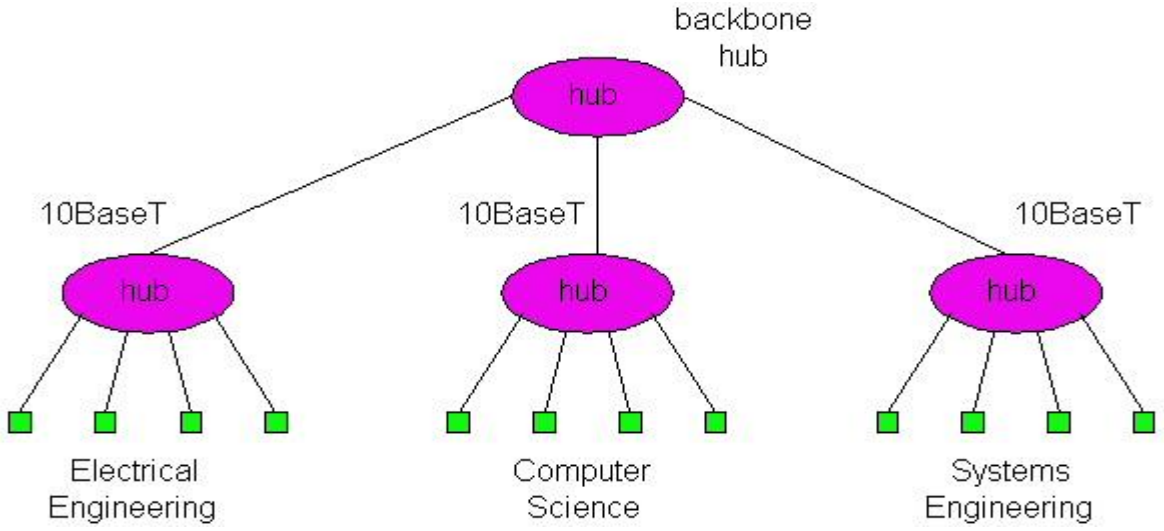
Interconnecting LANs

Q: Why not just one big LAN?

- Limited amount of supportable traffic: on single LAN, all stations must share bandwidth
- large "collision domain" (can collide with many stations)
- limited length: 802.3 specifies maximum cable length

Hubs

- Physical Layer devices: essentially repeaters operating at bit levels: repeat received bits on one interface to all other interfaces
- Hubs can be arranged in a hierarchy (or multi-tier design), with backbone hub at its top



Hubs (more)

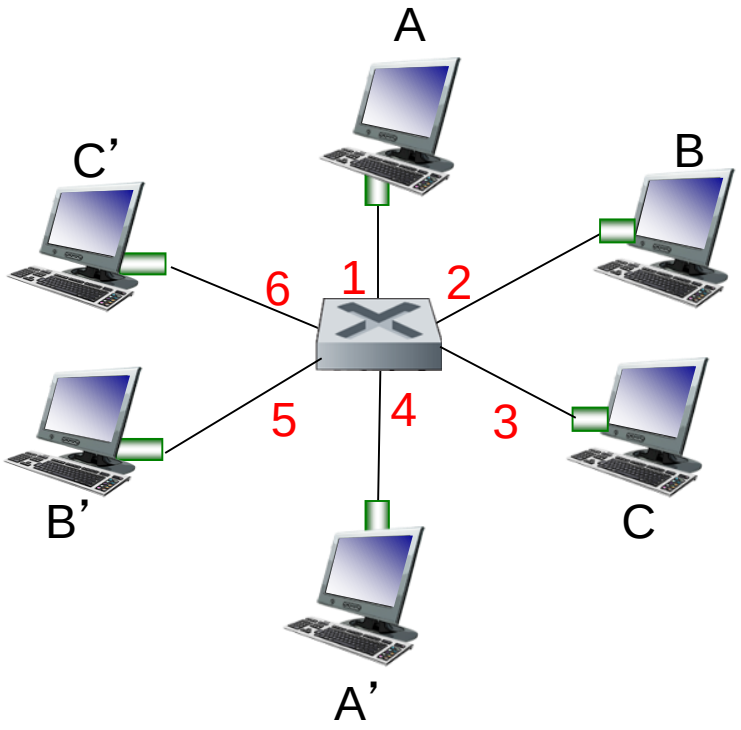
- Each connected LAN referred to as LAN **segment**
- Hubs do not isolate collision domains: node may collide with any node residing at any segment in LAN
- Hub Advantages:
  - simple, inexpensive device
  - Multi-tier provides graceful degradation: portions of the LAN continue to operate if one hub malfunctions
  - extends maximum distance between node pairs (100 m per Hub)

Hub limitations

- single collision domain results in no increase in max throughput
  - multi-tier throughput same as single segment throughput
- individual LAN restrictions pose limits on number of nodes in same collision domain and on total allowed geographical coverage
- cannot connect different Ethernet types (e.g., 10BaseT and 100baseT -- Twisted-pair cables for distances up to 100 m providing different speed (i.e., 10 Gbps and 100 Gbps, resp.)

Switch: *multiple* simultaneous transmissions

- hosts have dedicated, direct connection to switch
- switches buffer packets
- Ethernet protocol used on *each* incoming link, but no collisions; full duplex
  - each link is its own collision domain
- *switching:* A-to-A' and B-to-B' can transmit simultaneously, without collisions



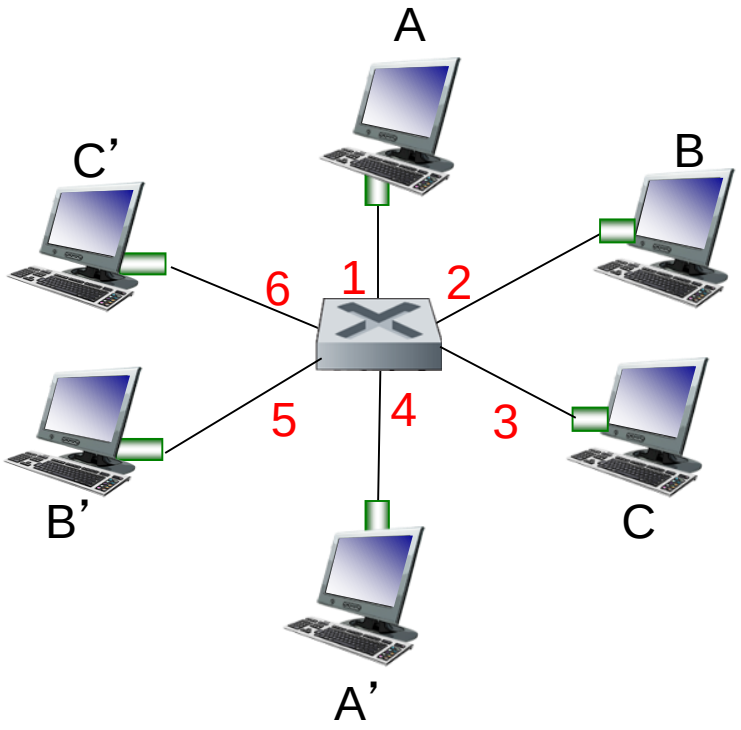
switch with six interfaces (1,2,3,4,5,6)

Ethernet switch

- link-layer device: takes an *active* role
  - store, forward Ethernet frames
  - examine incoming frame's MAC address, selectively forward frame to one-or-more outgoing links when frame is to be forwarded on segment, uses CSMA/CD to access segment
- *transparent*
  - hosts are unaware of presence of switches
- *plug-and-play, self-learning*
  - switches do not need to be configured

Switch: *multiple* simultaneous transmissions

- hosts have dedicated, direct connection to switch
- switches buffer packets
- Ethernet protocol used on *each* incoming link, but no collisions; full duplex
  - each link is its own collision domain
- *switching:* A-to-A' and B-to-B' can transmit simultaneously, without collisions



switch with six interfaces (1,2,3,4,5,6)

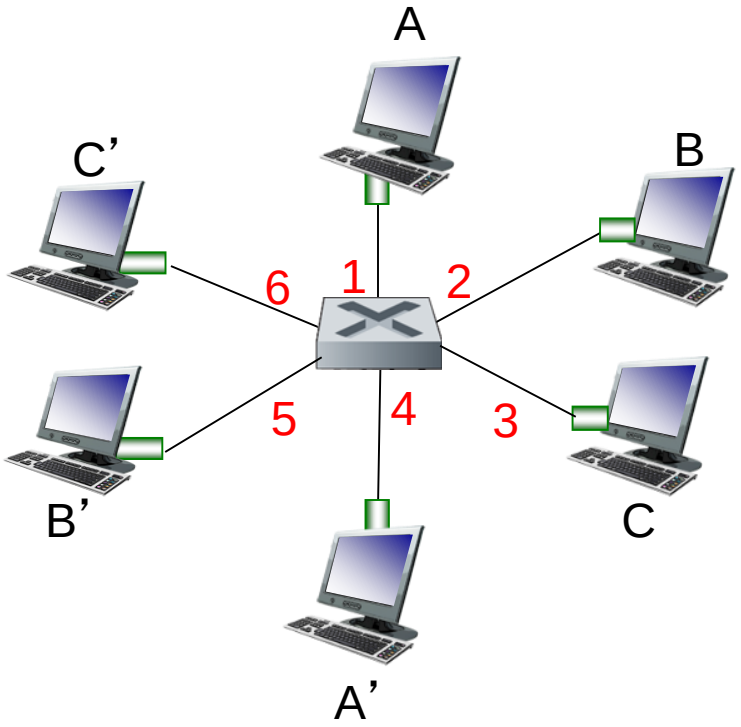
Switch forwarding table

Q: how does switch know A' reachable via interface 4, B' reachable via interface 5?

- A: each switch has a switch table, each entry:
  - (MAC address of host, interface to reach host, time stamp)
  - looks like a routing table!

Q: how are entries created, maintained in switch table?

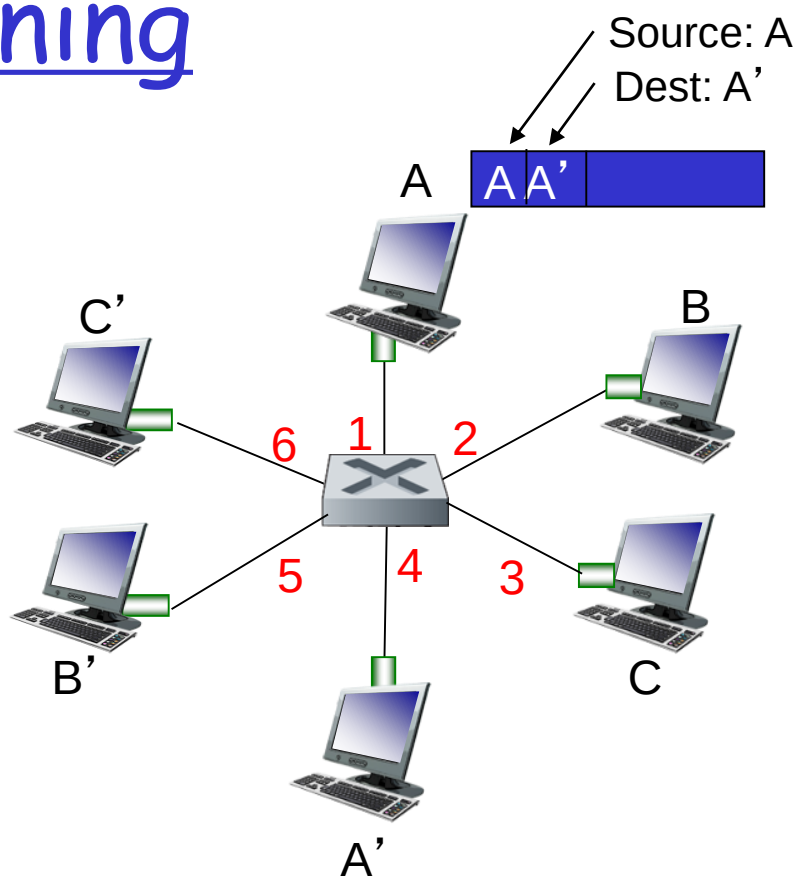
- something like a routing protocol?



switch with six interfaces (1,2,3,4,5,6)

Switch: self-learning

- switch *learns* which hosts can be reached through which interfaces
  - when frame received, switch "learns" location of sender: incoming LAN segment
  - records sender/location pair in switch table



MAC addr	interface	TTL
A	*1*	*60*

Switch table (initially empty)

Switch: frame filtering/forwarding

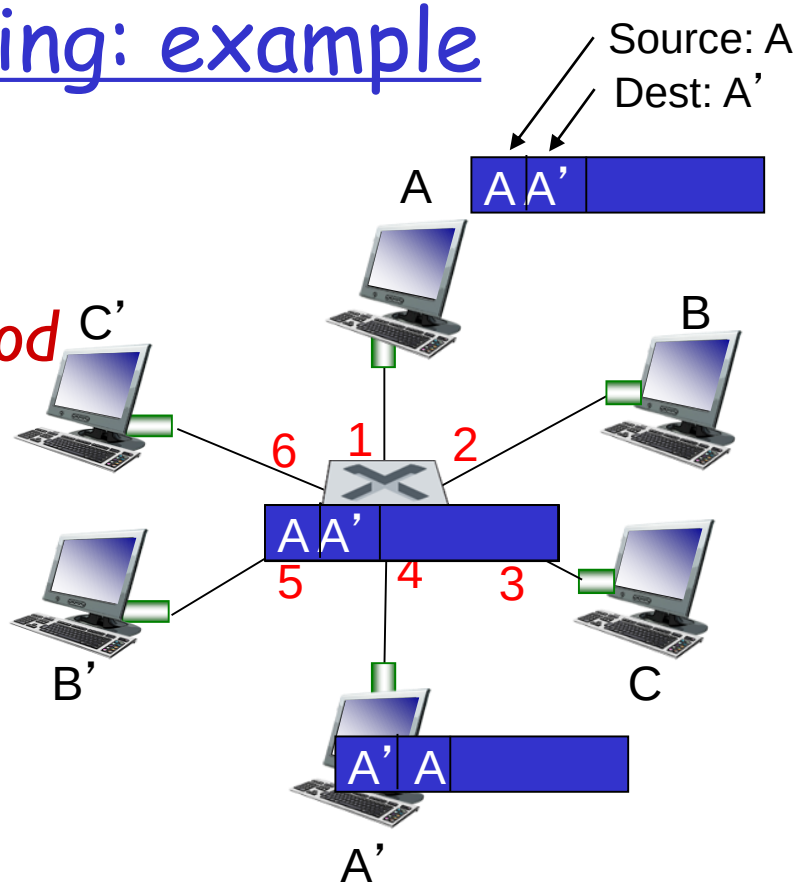
when frame received at switch:

1. record incoming link, MAC address of sending host
2. index switch table using MAC destination address
3. if entry found for destination
   then {
   if destination on segment from which frame arrived
   then drop frame
   else forward frame on interface indicated by entry
   }
   else flood /* forward on all interfaces except arriving
   interface */

Self-learning, forwarding: example

- frame destination, A', location unknown: *flood*
- destination A location known:

  selectively send on just one link

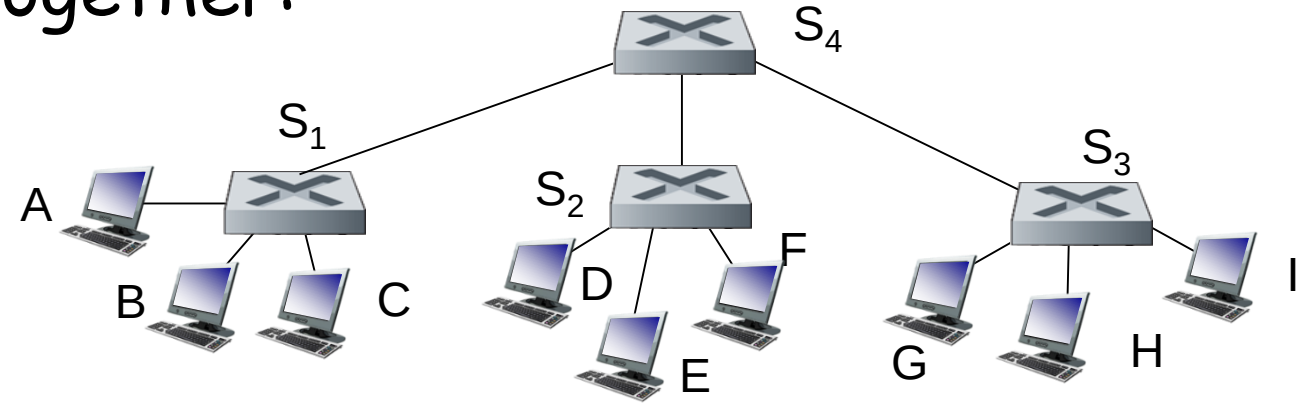


MAC addr	interface	TTL
A	*1*	*60*
A'	*4*	*60*

switch table (initially empty)

Interconnecting switches

self-learning switches can be connected together:

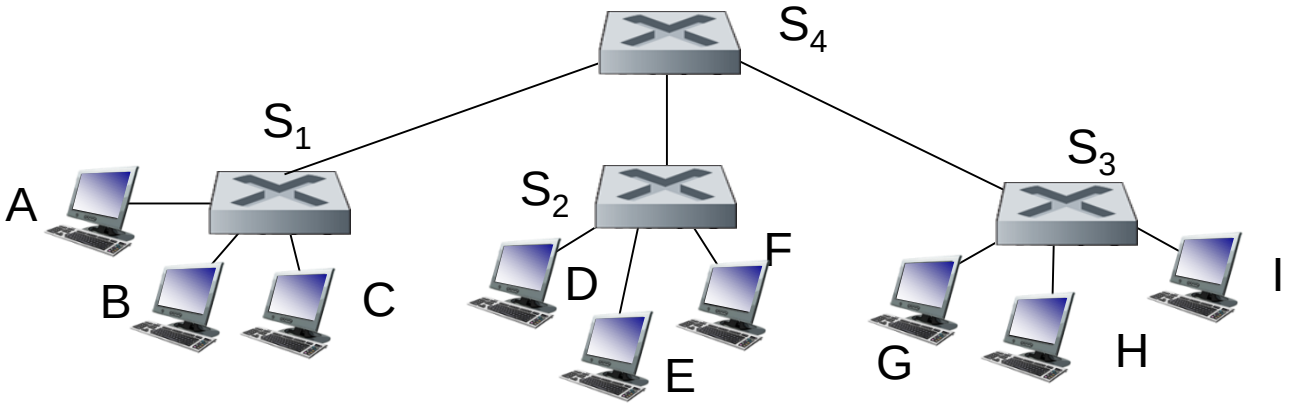


Q: sending from A to G - how does S_1 know to forward frame destined to G via S_4 and S_3?

- A: self learning! (works exactly the same as in single-switch case!)

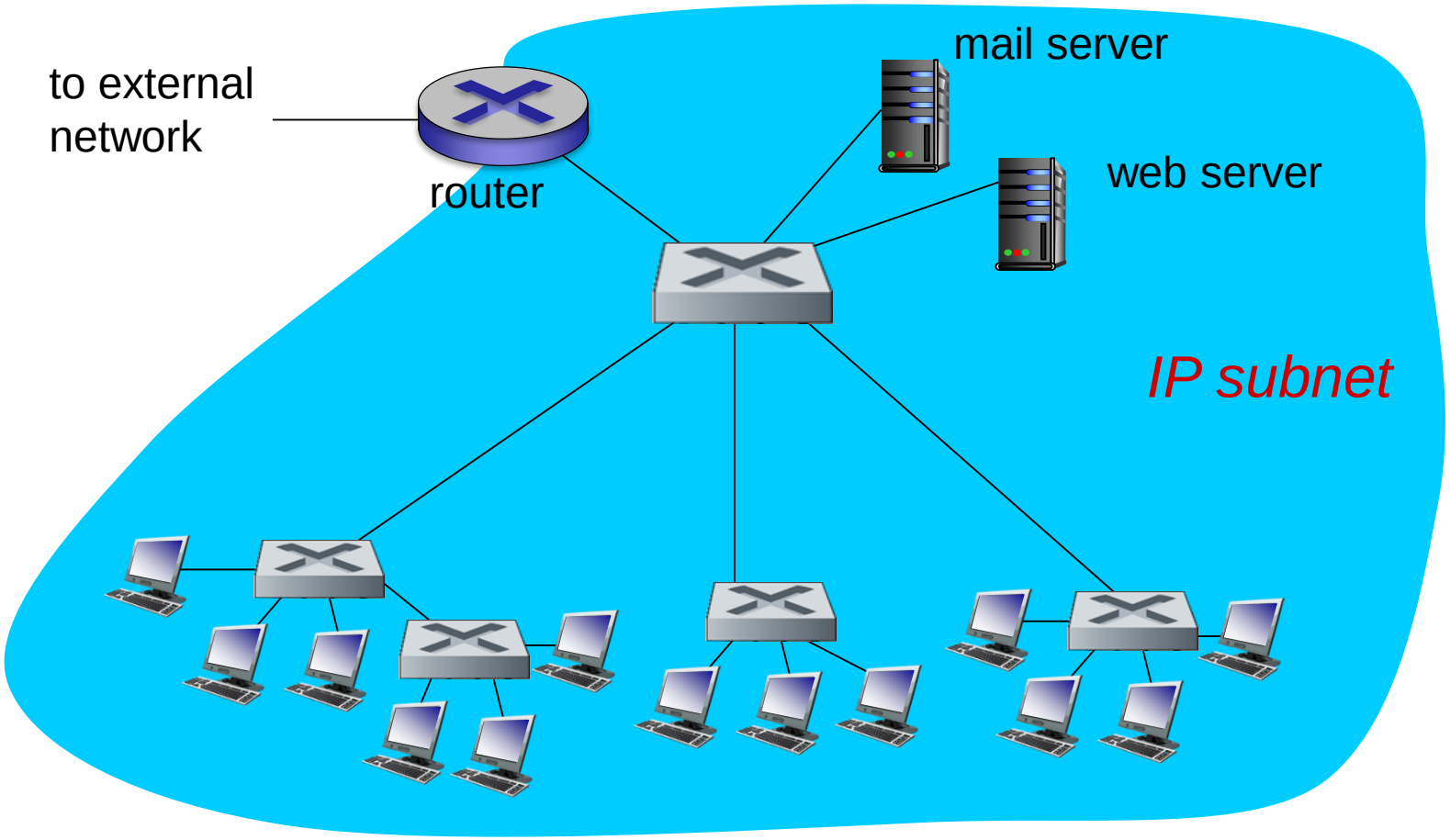
Self-learning multi-switch example

Suppose C sends frame to I, I responds to C

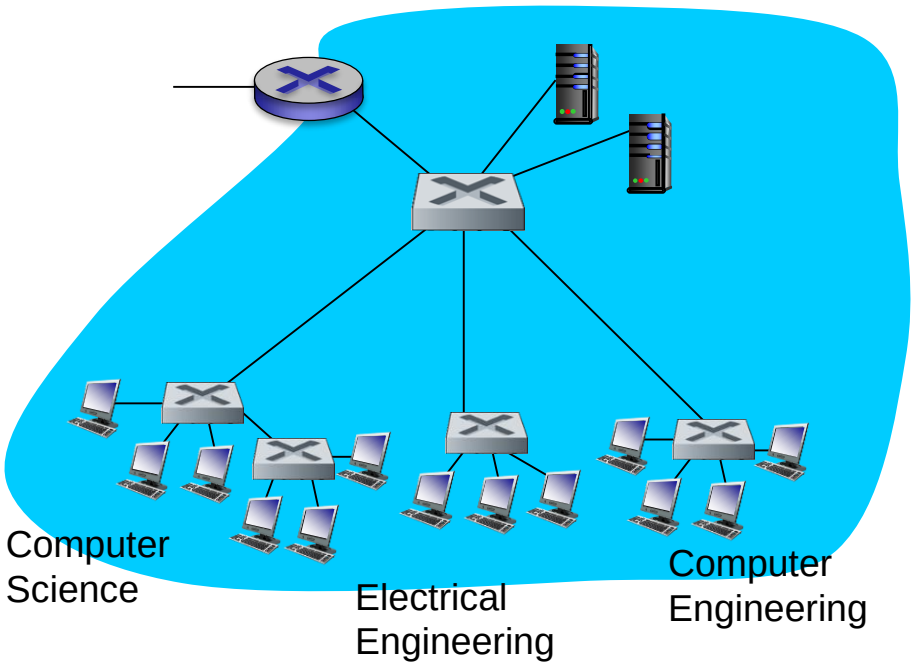


- Q: show switch tables and packet forwarding in S_1, S_2, S_3, S_4

Institutional network



VLANs: motivation



consider:

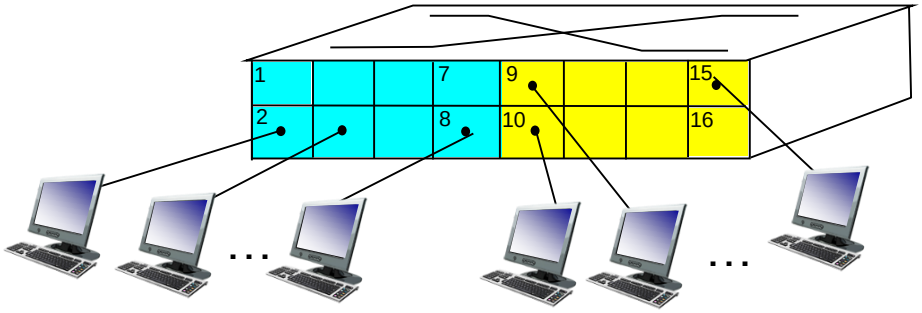
- CS user moves office to EE, but wants connect to CS switch?
- single broadcast domain:
  - all layer-2 broadcast traffic (ARP, DHCP, unknown location of destination MAC address) must cross entire LAN
  - security/privacy, efficiency issues

VLANs

Virtual Local Area Network

switch(es) supporting VLAN capabilities can be configured to define multiple ***virtual*** LANS over single physical LAN infrastructure.

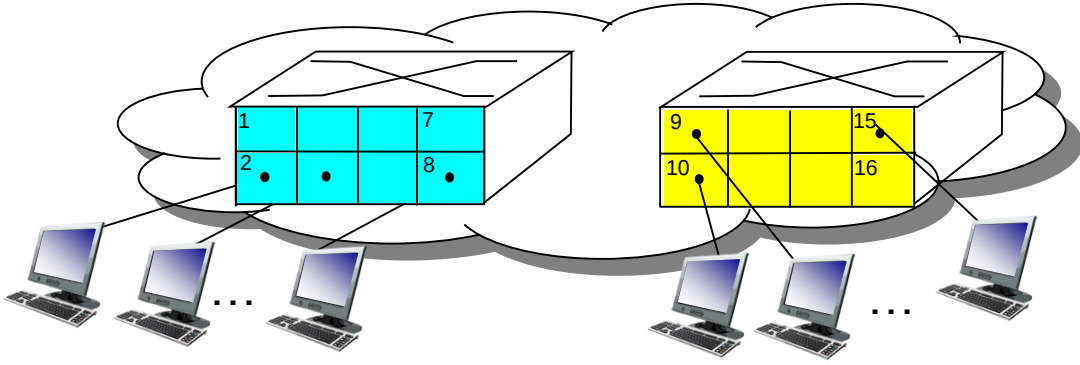
port-based VLAN: switch ports grouped (by switch management software) so that *single* physical switch ……



Electrical Engineering (VLAN ports 1-8)

Computer Science (VLAN ports 9-15)

… operates as multiple virtual switches

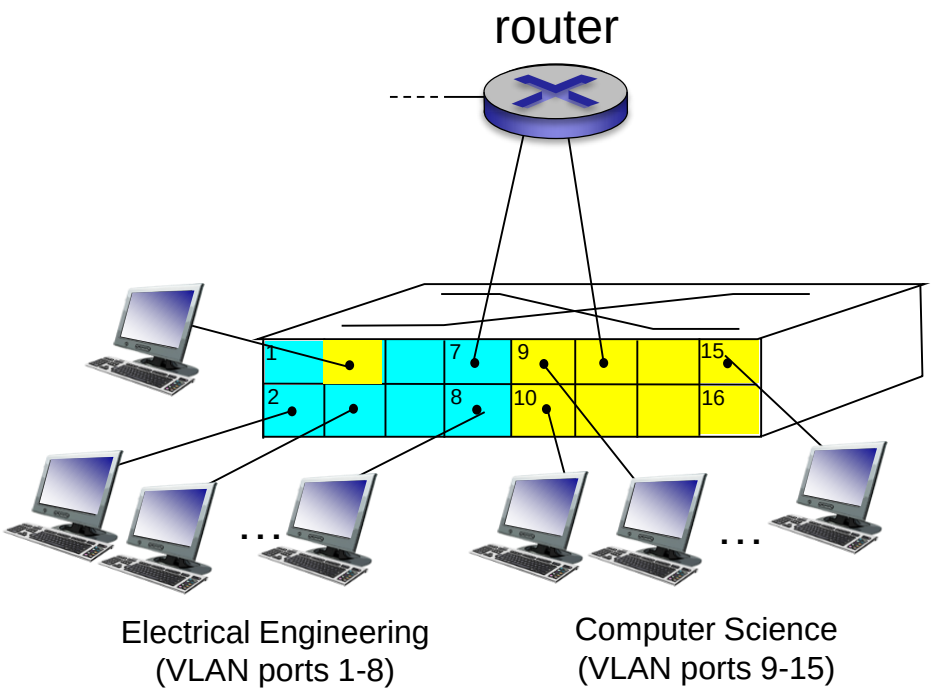


Electrical Engineering (VLAN ports 1-8)

Computer Science (VLAN ports 9-16)

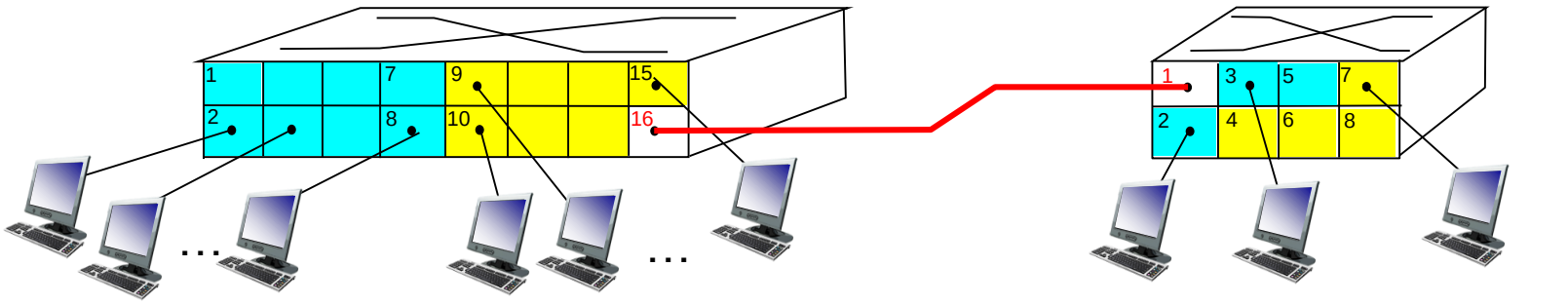
Port-based VLAN

- *traffic isolation:* frames to/from ports 1-8 can *only* reach ports 1-8
  - can also define VLAN based on MAC addresses of endpoints, rather than switch port
- dynamic membership: ports can be dynamically assigned among VLANs



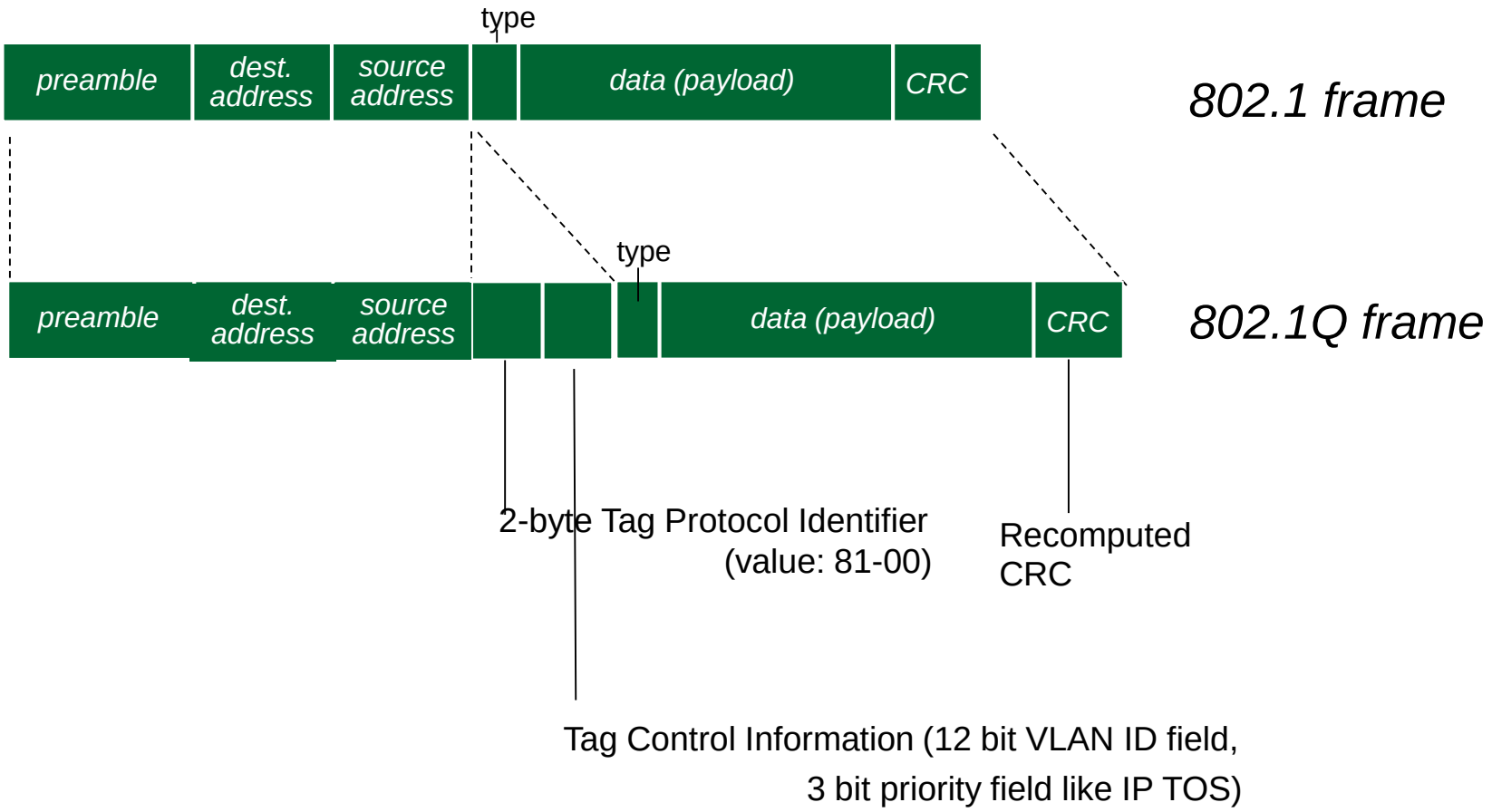
- forwarding between VLANS: done via routing (just as with separate switches)
  - in practice vendors sell combined switches plus routers

VLANS spanning multiple switches



- *trunk port:* carries frames between VLANS defined over multiple physical switches
  - frames forwarded within VLAN between switches can't be vanilla 802.1 frames (must carry VLAN ID info)
  - 802.1q protocol adds/removed additional header fields for frames forwarded between trunk ports

802.1Q VLAN frame format



Chapter 4:Local area network Summary

- various link layer technologies
  - LAN model
  - Ethernet
  - hubs, switches
  - *VLAN